

# Stability

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## Software Engineering/Mechatronics 3DX4

### Slides 6: Stability

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Material based on lecture notes by P. Taylor and M. Lawford, and *Control Systems Engineering* by N. Nise.

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## Introduction

- ▶ In this Section, we examine ways to determine if a system is **stable**.
- ▶ Of all design criteria, stability is most important.
- ▶ If system is unstable, then transient response and steady-state

error are irrelevant.

- ▶ We will now examine a few stability definitions for linear, time-invariant systems.

## Stability and Natural Response

- ▶ The total response of a system is

$$c(t) = c_{forced}(t) + c_{natural}(t)$$

1. A system is **stable** if natural response tends to zero as  $t \rightarrow \infty$ .
  2. A system is **unstable** if natural response grows unbounded as  $t \rightarrow \infty$ .
  3. A system is **marginally stable** if natural response neither decays or grows (eg. stays constant or oscillates with fixed amplitude) as  $t \rightarrow \infty$ .
- ▶ Definition of “stable” implies that as  $t \rightarrow \infty$ , only the forced response remains.

## Bounded-input, Bounded-output (BIBO) Stability

- ▶ The BIBO definition is in terms of the total response, so you don't need to isolate the natural response first.
- 1. A system is **stable** if every bounded input produces a bounded output.
- 2. A system is **unstable** if *any* bounded input produces an unbounded output.

## Stability and Poles

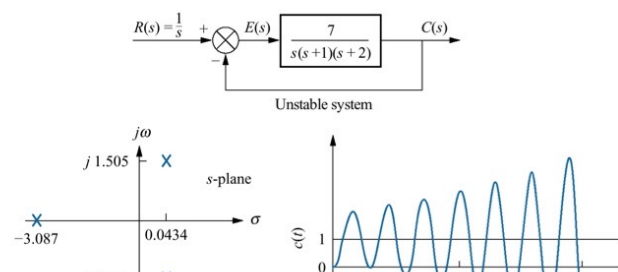
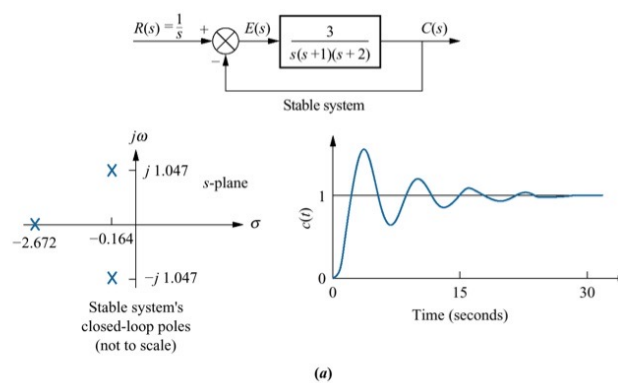
- ▶ In order to easily determine if a system is stable, we can examine the poles of the closed-loop system.
- 1. A system is **stable** if all the poles are strictly on the left hand side of the complex plane.
- 2. A system is **unstable** if any pole is in the right hand side of the complex plane or the system has poles on the imaginary axis that are of multiplicity  $> 1$ .
- 3. A system is **marginally stable** if no pole is on the right hand side, and its poles on the imaginary axis are of multiplicity one.

ie.  $\frac{1}{(s^2 + \omega^2)}$  marginally stable, but  $\frac{1}{(s^2 + \omega^2)^2}$  is not.

## Stability and Poles - II

- Poles on the imaginary axis of multiplicity greater than one have time responses of the form  $At^n \cos(\omega t + \phi)$  which tend to infinity as  $t \rightarrow \infty$ .
- This implies systems with poles on the imaginary axis of multiplicity one will be unstable by the BIBO definition as a sinusoid input at same frequency ( $\omega$ ) will result in a total response with imaginary poles of multiplicity two!

## Stability and Poles - III



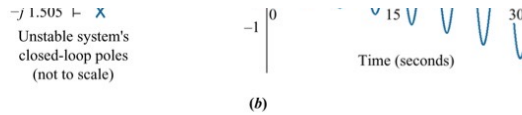


Figure 6.1.

## Stability Summary

**Table:** Stability Comparison

| Real Part of Poles                                      | Natural Response  | BIBO     |
|---------------------------------------------------------|-------------------|----------|
| All poles $< 0$                                         | stable            | stable   |
| Any pole $> 0$ or imaginary poles of multiplicity $> 1$ | unstable          | unstable |
| Poles $\leq 0$ and imaginary poles of multiplicity one  | marginally stable | unstable |

他俩的区别是：natural response描述的是系统在没有输入的时候，内部自然响应会不会自然衰减，而BIBO描述的是有界的输入会不会产生有界的输出。

**注意：**最后一个条件说这里说的虚轴上的pole是必须单根，即s的次数是1。

## Closed-loop Systems

- If the poles of the original system are not as desired, we can use feedback control to move the poles.

use feedback control to move the poles.

- ▶ In Fig. 6.2(a) below, we can easily see the poles of original system, but we don't know the poles of closed-loop system without factoring.
- ▶ Would like an easy way to tell if the closed-loop system is stable without having to factor it.

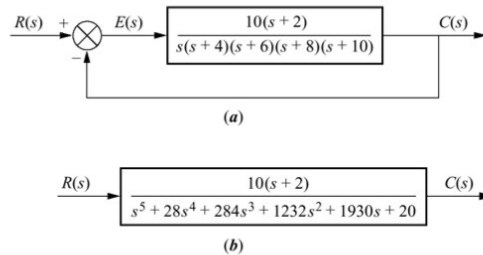


Figure 6.2.

## Necessary Stability Condition

- ▶ A *necessary* condition for a polynomial to have all roots in the open left hand plane is to have all coefficients of the polynomial to be present and to have the same sign.
- ▶ However, this is not a *sufficient* condition.
- ▶ A *sufficient* condition that a system is *unstable* is that all coefficients do not have the same sign.
- ▶ If some coefficients are missing, system MAY be unstable, or at best, marginally stable.
- ▶ If all coefficients are same sign and present, system could be stable or unstable.

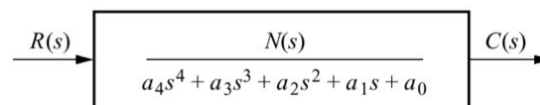


Figure 6.3.

这里说的是：如果 $G(s)$ 的分母系数是全的（就是比如说 $s^4, s^3$ 一直到 $s^0$ 的系数都不是0），那么观察其是不是同号，如果不同直接不稳定，如果相同去做RH表；缺项也要做RH表。

## Routh-Hurwitz Criterion

- ▶ This method will give us stability info without having to find poles of closed-loop system.
- ▶ Will tell us:
  - ▶ How many poles in left half-plane.
  - ▶ How many poles in right half-plane.
  - ▶ How many poles on imaginary axis.
- ▶ Method called **Routh-Hurwitz criterion** for stability.
- ▶ To apply method we need to:
  1. Construct a table of data called a *Routh table*.
  2. Interpret the table to determine the above classifications.

## Creating a Basic Routh Table

- ▶ The Routh-Hurwitz criterion focusses on the coefficients of the denominator of the transfer function.
- 1. Label the rows of the table with powers of  $s$ , starting from the highest power down to  $s^0$ .
- 2. List coefficients across top row, starting with coefficient of the highest power of  $s$ , and then every other coefficient.



3. List remaining coefficients in second row, starting with coefficient of second highest power.

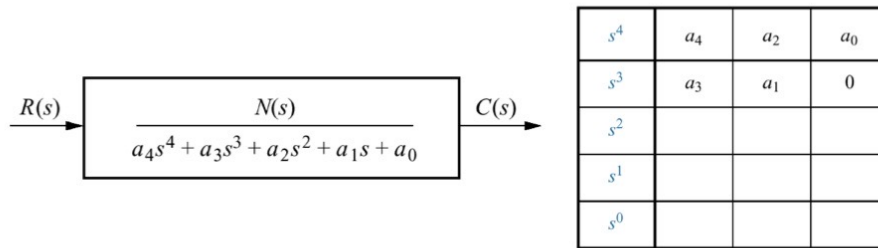


Figure 6.3 and Table 6.1.

**注意：RH table的列数是动态的，我们需要把characteristic polynomial的所有系数在前两行写下，根据这个调整列数。**

**行数是由最高系数决定的，最高n次项就要n+1列。**

## Creating a Basic Routh Table - II

► To fill in remaining rows as follows:

1. Each entry is a negative determinant of entries from the previous two rows.
2. Each determinant is divided by the entry in the first column of the row above.
3. Left column of determinant is first column of the previous two rows.
4. Right column contains elements of the column above and directly to the right of the current location.
5. If no column to right, use zeros.

|       |                                                                           |                                                                         |                                                                     |
|-------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| $s^4$ | $a_4$                                                                     | $a_2$                                                                   | $a_0$                                                               |
| $s^3$ | $a_3$                                                                     | $a_1$                                                                   | 0                                                                   |
| $s^2$ | $-\frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$ | $-\frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$ | $-\frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$ |
| $s^1$ | $-\frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$ | $-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$     | $-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ |
| $s^0$ | $-\frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$   | $-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$     | $-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ |

Table 6.2



简单来讲，就是第一列sign change的次数代表右半平面pole的数量，可以靠这个来判断stability。

## Interpreting a Basic Routh Table

- ▶ Basic Routh table applies to systems with poles in open left or right hand plane, but no imaginary poles.
- ▶ The **Routh-Hurwitz criterion** states that the number of poles in the right half plane is equal to the number of sign changes in the first coefficient column of the table.
- ▶ A system is **stable** if there are no sign changes in the first column.

## Basic Routh Table eg.

- ▶ Apply the Routh-Hurwitz criterion to the system below to determine stability.
- ▶ One may multiply any row by a **positive** constant without changing the values of the rows below.

- You MUST not multiple a row by a **negative** constant.

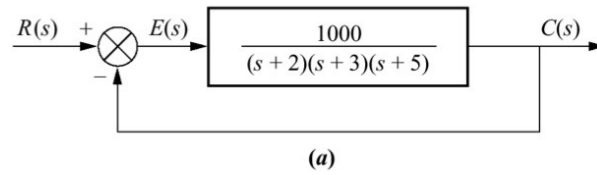


Figure 6.4.

**注意：RH 表拿close loop transfer function  $1+G(s)H(s)=0$ 去做，而不是 open loop  $G(s)$ ，要注意分清！**

**做表的时候每一行可以任意乘除一个非负数。**

Ex 1.

$$G(s) = \frac{1000}{(s+2)(s+3)(s+5)} = \frac{1000}{s^3 + 10s^2 + 31s + 30}$$

$$G_{CL}(s) = \frac{G}{1+GH}, \quad H=1$$

$$= \frac{1000}{s^3 + 10s^2 + 31s + 1030}$$

|       |                                                             |                                                      |
|-------|-------------------------------------------------------------|------------------------------------------------------|
| $s^3$ | 1                                                           | 31                                                   |
| $s^2$ | 10 $\Rightarrow$   sign changes                             | 1030 $\Rightarrow$ 103                               |
| $s^1$ | $-\begin{vmatrix} 1 & 31 \\ 10 & 1030 \end{vmatrix} = -72$  | $-\begin{vmatrix} 1 & 0 \\ 10 & 0 \end{vmatrix} = 0$ |
| $s^0$ | $-\begin{vmatrix} 10 & 1030 \\ -72 & 0 \end{vmatrix} = 103$ | 0                                                    |

2 poles in RHP (2 sign changes), 1 pole in LHP (3 total)

Can  $\times / \div$  any constant by any row.

## Case I: Zero Only in First Column - $\epsilon$

- Consider system with closed-loop transfer function:

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3} \quad (1)$$

- Replace zero entry in first column by an  $\epsilon$  (very small  $\rightarrow \epsilon$  number), then complete table.
- Examine table by allowing  $\epsilon$  to approach zero from the positive and negative side.

|       | Label      | First Column                     | $\epsilon = +$ | $\epsilon = -$ |
|-------|------------|----------------------------------|----------------|----------------|
| $s^5$ |            | 1                                | +              | +              |
| $s^4$ |            | 2                                | +              | +              |
| $s^3$ | $\epsilon$ | $\epsilon$                       | +              | -              |
| $s^2$ |            | $\frac{6\epsilon - 7}{\epsilon}$ | -              | +              |
| $s^1$ |            | $42\epsilon - 49 - 6\epsilon^2$  | +              | +              |

|       |                   |       |       |
|-------|-------------------|-------|-------|
| $s^5$ | $12\epsilon - 14$ | $s^5$ | $s^5$ |
| $s^0$ | 3                 | $s^0$ | 3     |
|       |                   |       | +     |
|       |                   |       | +     |

Tables 6.4 and 6.5.

如果我们在第一列出现了0，那么我们把它用一个很小的数epsilon替代，然后代入计算，之后我们分别假设从正和负侧逼近，即epsilon为+或者-，看看第一列怎么变。

- 如果两边都没有符号变化 → 没有右半平面根
- 如果有  $m$  次符号变化 → 有  $m$  个右半平面根
- 如果正负两边结果不同，说明有根落在虚轴附近，需要特别小心

## Case I: Zero Only in First Column - Reciprocal

- ▶ A polynomial whose roots are the reciprocal of the original polynomial, has poles with same distributions (ie. # in left side, right side, imaginary).
- ▶ This new polynomial **might** not have a zero in the first column.
- ▶ Can find this polynomial simply by **reversing order of coefficients.**

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3} \rightarrow D(s) = \frac{10}{1 + 2s^1 + 3s^2 + 6s^3 + 5s^4 + 3s^5}$$

$$D(s) = \frac{10}{3s^5 + 5s^4 + 6s^3 + 3s^2 + 2s^1 + 1}$$

|       |       |     |   |
|-------|-------|-----|---|
| $s^5$ | 3     | 6   | 2 |
| $s^4$ | 5     | 3   | 1 |
| $s^3$ | 4.2   | 1.4 |   |
| $s^2$ | 1.33  | 1   |   |
| $s^1$ | -1.75 |     |   |
| $s^0$ | 1     |     |   |

2 poles in RHP

Table 6.6.

第一步为构造新的多项式系数表，然后进行判断，因为反转后的多项式和原多项式：

- 左半平面根个数相同
- 右半平面根个数相同
- 虚轴根个数相同

## Case II: Row of Zeros

- Consider closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56} \quad (2)$$

- When evaluating row  $s^3$ , we find all entries to be zero.
- To proceed, form polynomial using coefficients of row above the zero row.
- Start with power of row above the zero row, and then skip every other power of  $s$ .
- This gives us:

$$P(s) = s^4 + 6s^2 + 8 \quad (3)$$

- Next, differentiate with respect to  $s$

$$\frac{dP(s)}{ds} = 4s^3 + 12s^1 + 0 \quad (4)$$

|       |                             |                              |                             |
|-------|-----------------------------|------------------------------|-----------------------------|
| $s^5$ | 1                           | 6                            | 8                           |
| $s^4$ | <del>7</del> 1              | <del>42</del> 6              | <del>56</del> 8             |
| $s^3$ | <del>8</del> <del>4</del> 1 | <del>8</del> <del>12</del> 3 | <del>8</del> <del>8</del> 0 |
| $s^2$ | 3                           | 8                            | 0                           |
| $s^1$ | $\frac{1}{3}$               | 0                            | 0                           |
| $s^0$ | 8                           | 0                            | 0                           |

÷7

如果出现一整行是0，那么我们先看上一行，写出对应的polynomial  $A(s)$ ，然后对 $s$ 求一次导，然后找到新的多项式对应的系数重新分配给这一行。

全零行说明有对称的pole，我们用上一步得到的 $A(s)$ 求解，看看pole在哪。

如果我们是用reverse的方法做的RH table，那么用 $A$ 找根的时候用的是求解出来的root的倒数。

## Case II: Row of Zeros - II

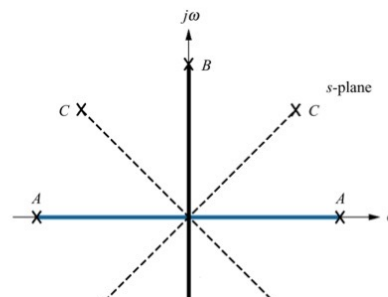
- Replace row of zeros with coefficients of polynomial from equation 4, and continue.

Table 6.7

|       |               |              |              |
|-------|---------------|--------------|--------------|
| $s^5$ | 1             | 6            | 8            |
| $s^4$ | <del>1</del>  | <del>6</del> | <del>8</del> |
| $s^3$ | <del>1</del>  | <del>3</del> | <del>0</del> |
| $s^2$ | 3             | 8            | 0            |
| $s^1$ | $\frac{1}{3}$ | 0            | 0            |
| $s^0$ | 8             | 0            | 0            |

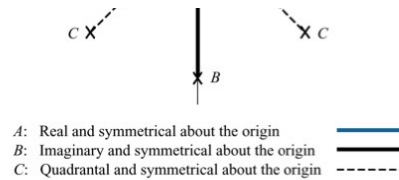
## Why Row of Zeros?

- We get a row of zeros when original polynomial has a purely even or odd polynomial as a factor.
  - A purely even polynomial is one where all powers of  $s$  are even.
  - An even polynomial only has roots that are symmetrical about origin.
  - As  $j\omega$  roots are symmetric across origin, they can only occur when we have a row of zeros.
- In Routh table, the row above the row of zeros contains the even/odd polynomial that is a factor of the



original polynomial.

Figure 6.5



## Why Row of Zeros? - II

- ▶ Also, Everything from row containing even polynomial onwards is a test of *only* the even polynomial.
- ▶ Returning to our example with  $T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56}$ , we see that it had even polynomial  $P(s) = s^4 + 6s^2 + 8$  as a factor.
- ▶ Rows  $s^4$  to  $s^0$  thus give information only about  $P(s)$ .
- ▶ As there are no sign changes, we thus have 4 imaginary poles.
- ▶ As  $P(s)$  is not perfect fourth order square polynomial, the imaginary poles are of multiplicity 1.
- ▶ There are no sign changes from rows  $s^5$  to  $s^4$ , so our last pole is in left hand side.

- ▶ System is thus marginally stable.

Table 6.7

|       |               |    |    |
|-------|---------------|----|----|
| $s^5$ | 1             | 6  | 8  |
| $s^4$ | 7             | 42 | 56 |
| $s^3$ | 6             | 42 | 0  |
| $s^2$ | 3             | 8  | 0  |
| $s^1$ | $\frac{1}{3}$ | 0  | 0  |
| $s^0$ | 8             | 0  | 0  |

当传递函数的分母出现了纯偶多项式，比如说这里的 $P(s)$ ，那么说明其产生的root关于原点对称。比如说如果有纯虚数根，说明一定有一整行是0的。

**注意！！全0行只说明此时系统有一对pole是关于原点对称的，不一定是marginally stable！除非是一对纯虚数pole。**

在这个例子里面，我们在第三行遇到了全0，于是我们找到上一行，并写成



polynomial是 $A(s)=7s^4+42s^2+56$ ，同时这一行可以同时除以7，然后对 $A(s)$ 求导，得到 $28s^3+84s$ ，化简得到 $1s^3+3s$ 。于是1, 3, 0就是第三行的参数。

出现全0行，我们直接对 $A(s)=0$ 求解，看看 $s$ 是不是纯虚数根。

### 判断方法：

#### 1. Stable

如果第一列没有sign change，没有出现整行0，第一列没有0，那么就是stable。

#### 2. Unstable

如果第一列出现符号变化，那么变化的次数就是RHP的pole个数，所以unstable。

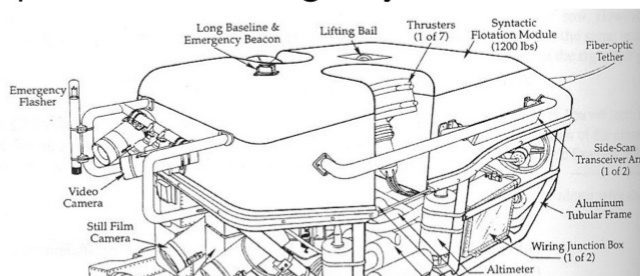
#### 3. Marginal stable

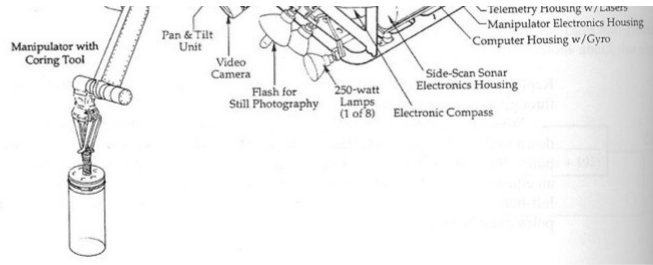
表里出现整行0，同时第一列又没有符号变化。

如果我们同时出现整行0和第一列0，那么谁先出现就先处理谁；如果它们发生在同一行，优先处理整行0的情况。

## Stability Design via Routh-Hurwitz

- Changes in the gain of systems like the one below, can result in changes of the closed-loop pole locations.
- In the next example, we can use the Routh-Hurwitz criterion to show that gain changes can move stable poles from the right-hand plane, to the imaginary axis, to the left-hand plane.





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Figure 6.9.

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## Stability Design via Routh-Hurwitz eg.

- Find range of  $K$  (gain) that will make the system stable, marginally stable, and unstable.

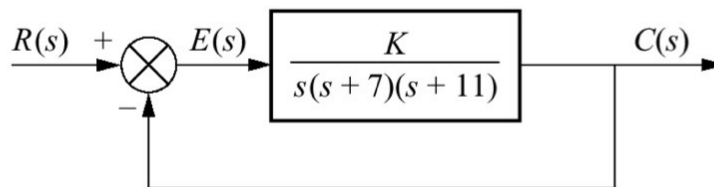


Figure 6.10.

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$$G(s) = \frac{k}{s(s+7)(s+11)} \quad H(s) = 1$$

$$T(s) = \frac{G}{1+GH} = \frac{k}{s(s+7)(s+11) + k}$$

$$= s^3 + 18s^2 + 77s + k$$

|       |                     |    |
|-------|---------------------|----|
| $s^3$ | 1                   | 77 |
| $s^2$ | 18                  | k  |
| $s^1$ | $\frac{1386-k}{18}$ | 0  |
| $s^0$ | k                   | 0  |

$$\frac{- \begin{vmatrix} 1 & 77 \\ 18 & k \end{vmatrix}}{18} = \frac{18(77) - k}{18} \quad - \begin{vmatrix} 1 & 0 \\ 18 & 0 \end{vmatrix} = 0$$

$$= \frac{1386 - k}{18}$$

$$\frac{- \begin{vmatrix} 18 & k \\ \frac{1386-k}{18} & 0 \end{vmatrix}}{\frac{1386-k}{18}} = k \quad 0$$

For stability, no sign changes

$$\frac{1386 - k}{18} > 0 \text{ from } s^1 \text{ row} \Rightarrow k < 1386$$

$$k > 0 \text{ from } s^0 \text{ row}$$

$$\text{This means } 0 < k < 1386$$

## Stability in State Space

- For a state space system, we are given the state and output equations below

$$\dot{\underline{x}} = \underline{A}\underline{x} + \underline{B}\underline{u} \quad \text{state equations}$$

$$\underline{y} = \underline{C}\underline{x} + \underline{D}\underline{u} \quad \text{output equations}$$

- If we have a single input, single output system, we can use the equation below to find the corresponding transfer function. :

$$G(s) = \frac{Y(s)}{U(s)} = \underline{C}(s\underline{I} - \underline{A})^{-1}\underline{B} + \underline{D} \quad (5)$$

If it has multiple inputs or outputs then we get  $\underline{G}(S)$ , a TF matrix.

- From linear algebra we know:

$$[s\underline{I} - \underline{A}]^{-1} = \frac{\text{adj}([s\underline{I} - \underline{A}])}{\det([s\underline{I} - \underline{A}])} \quad (6)$$

## Stability in State Space - II

- ▶ Substituting equation 6 into equation 5, we get

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\underline{C} \operatorname{adj}([s\underline{I} - \underline{A}]) \underline{B}}{\det([s\underline{I} - \underline{A}])} + \underline{D} = \frac{N(s)}{D(s)} \quad (7)$$

$$= \frac{\underline{C} \operatorname{adj}([s\underline{I} - \underline{A}]) \underline{B} + \det([s\underline{I} - \underline{A}]) \underline{D}}{\det([s\underline{I} - \underline{A}])} \quad (8)$$

- ▶ We thus have:

$$\det([s\underline{I} - \underline{A}]) = D(s) \quad (9)$$

- ▶ We define the roots of the equation  $\det([s\underline{I} - \underline{A}]) = 0$  to be the **eigenvalues** of matrix  $\underline{A}$ .
- ▶ To determine if a state space system is stable, we determine the eigenvalues of matrix  $\underline{A}$ , and then determine their location in the  $s$ -plane, using the same rules for stability as for the poles of a transfer function.

简单来说就是，如果用state space来表示一个系统的传递函数G，那么G的pole取决于 $\det(sI - A)$ ，也就是A的eigenvalue，在频域上的表现。根据pole的位置来判断stability。

### 做题步骤：

1. 如果是给出了open loop  $G(s)$ ，那么写出 $1 + H(s) * G(s) = 0$ 作为characteristic polynomial，然后求出polynomial每一项的多项式系数，填入RH表里面。如果是state space，那么写出 $\det[sI - A]$ ，让其=0

得出特征多项式。

2. 如果特征多项式的所有系数都不是0，那么观察，如果所有系数（包括常数项）都是相同符号，那么继续判断，如果有不同的直接不稳定。
3. 如果第一列出现0，那么使用 $\epsilon$ 来代替0，或者把特征多项式的系数反转，叫reciprocal，但是记住，如果用了reciprocal那么确定是不是有虚数根的时候，要把 $A(s)$ 中的 $s$ 换成 $1/s$ 。
4. 如果出现整列0，那么把上一行的系数写成多项式 $A(s)$ ，然后对 $s$ 求导求出这一行的新系数。
5. 如果第一行出现符号变化，那么不稳定，变化几次就是几个RHP pole。如果出现全0行，那么有对称的pole，求解 $A(s)=0$ 来判断。如果有 $\epsilon$ ，那么 $\epsilon$ 为 $+-$ 的情况都要判断一次。

例子们：

16. For the unity-feedback system of [Figure P6.3](#) with

$$G(s) = \frac{K(s+3)(s+5)}{(s-2)(s-4)}$$

determine the range of  $K$  for stability. [[Section: 6.4](#)]

$$K > \frac{3}{4}; K < -1$$

17.  Use MATLAB and the Symbolic Math Toolbox to generate a Routh table in terms of  $K$  to solve [Problem 16](#).

怎么拿matlab写：

```
syms s
syms K real
% 1. 定义开环传递函数的分子和分母
num = K * (s + 3) * (s + 5);
den = (s - 2) * (s - 4);
% 2. 特征方程: 1 + G(s) = 0 => den + num = 0
char eq = expand(num + den);
```

```

char_eq = expand(num + den);
% 3. 提取特征方程中 s 的系数 (按降幂排列)
% 结果为一个向量 [s^2的系数, s^1的系数, s^0的系数]
c = coeffs(char_eq, s, 'All');
% 4. 建立劳斯表 (这是一个二阶系统, 可以直接手动填入前两行, 通过公式算第三行)
% 第一行: s^2 系数
% 第二行: s^1 系数
% 第三行: 第二行和第一行算出来的 s^0
RT = [c(1), c(3);
      c(2), 0];
% 计算第三行的元素
row3_col1 = simplify(-det([RT(1,1), RT(1,2); RT(2,1), RT(2,2)]) /
RT(2,1));
RT = [RT; row3_col1, 0];
disp('Routh Table:');
disp(RT);
% 5. 提取劳斯表的第一列
First_Column = RT(:, 1);
disp('Routh Table 第一列:');
disp(First_Column);

```

```

Routh Table:
[  K + 1, 15*K + 8]
[ 8*K - 6,      0]
[15*K + 8,      0]

Routh Table 第一列:
  K + 1
 8*K - 6
15*K + 8

```

**SS 2.** Tell how many roots of the following polynomial are in the right half-plane, in the left half-plane, and on the  $j\omega$ -axis: [\[Section: 6.3\]](#)

$$P(s) = s^5 + 6s^3 + 5s^2 + 8s + 20$$

Q2.



$p(s) = s^5 + 6s^4 + 5s^3 + 8s^2 + 20s + 20$ , note  $s^5$  in the first column, so reverse it

$$p(s) = 20s^5 + 8s^4 + 5s^3 + 6s^2 + 0s + 1$$

| $s_5$ | 20                                                                                          | 5                                                                                        | 0                                                                                |
|-------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| $s_4$ | 8                                                                                           | 6                                                                                        | 1                                                                                |
| $s_3$ | $-\left  \begin{array}{cc} 20 & 5 \\ 8 & 6 \end{array} \right $<br>8<br>$= -10$             | $-\left  \begin{array}{cc} 20 & 0 \\ 8 & 1 \end{array} \right $<br>8<br>$= \frac{-5}{2}$ | $-\left  \begin{array}{cc} 20 & 0 \\ 8 & 0 \end{array} \right $<br>8<br>$= 0$    |
| $s_2$ | $-\left  \begin{array}{cc} 8 & 6 \\ -10 & -\frac{5}{2} \end{array} \right $<br>-10<br>$= 4$ | $-\left  \begin{array}{cc} 8 & 1 \\ -10 & 0 \end{array} \right $<br>-10<br>$= 1$         | $-\left  \begin{array}{cc} 8 & 0 \\ -10 & 0 \end{array} \right $<br>-10<br>$= 0$ |
| $s_1$ | $-\left  \begin{array}{cc} -10 & -\frac{5}{2} \\ 4 & 1 \end{array} \right $<br>4<br>$= 0$ 8 | $-\left  \begin{array}{cc} -10 & 0 \\ 4 & 0 \end{array} \right $<br>4<br>$= 0$ 0         | 0 0                                                                              |
| $s_0$ | $-\left  \begin{array}{cc} 4 & 1 \\ 8 & 0 \end{array} \right $<br>8<br>$= 1$                | $-\left  \begin{array}{cc} 4 & 0 \\ 8 & 0 \end{array} \right $<br>8<br>$= 0$             | 0                                                                                |

$$s_1: 4s^2 + 1 \quad \frac{d}{ds}(4s^2 + 1) = 8s + 0$$

$$\text{let } 4s^2 + 1 = 0 \quad \therefore s^2 = -\frac{1}{4} \quad s = \pm \frac{1}{2}i$$

-

But we use reverse so:  $s' = \frac{1}{s} = \pm 2j$

$\therefore$  2 poles RHP, 2 poles on  $j\omega$  axis, 1 on LHP.

**35.** <sup>State Space</sup> **ss** The following system in state space represents the forward path of a unity-feedback system. Use the Routh–Hurwitz criterion to determine if the closed-loop system is stable. [[Section: 6.5](#)]

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ -5 & -4 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u \\ u &= [1 \ 0 \ 1] \mathbf{x}\end{aligned}$$

Q35.

$$\therefore G(s) = \frac{Y(s)}{U(s)} = \frac{C \operatorname{adj}[sI-A]B + \det[sI-A] \cdot D}{\det[sI-A]}$$

$\therefore$  check  $\det[sI-A]$

$$\therefore A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ -5 & -4 & -3 \end{bmatrix} \therefore \det[sI-A] = \begin{vmatrix} s & -1 & 0 \\ 0 & s-1 & -2 \\ 5 & 4 & s+3 \end{vmatrix}$$

$$= s[(s-1)(s+3) - (-2)4] + 1[0 - (-2)5] + 0$$

$$= s[s^2 + 2s - 3 + 8] + [10]$$

$$= s^3 + 2s^2 + 5s + 10$$

$$\therefore B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \therefore \text{only calculate } C_{21}, C_{22}, C_{23}$$

$$\operatorname{Adj}[sI-A] = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}, \quad C = [1 \ 0 \ 1], \quad D = 0$$

$$C_{21} = -[-(s+3) - 0] \\ = s+3$$

$$C_{22} = [s(s+3) - 0] = s^2 + 3s$$

$$C_{23} = -[4s+5] = -4s-5$$

$$\therefore C \operatorname{adj}[sI-A]B + \det[sI-A] \cdot D = [1 \ 0 \ 1] \cdot \begin{bmatrix} s+3 \\ s^2+3s \\ -4s-5 \end{bmatrix}$$

$$= s+3 - 4s-5$$

$$= -3s-2$$

$$\therefore G(s) = \frac{-3s-2}{s^3+2s^2+5s+10}$$

$$\therefore T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{-3s-2}{s^3+2s^2+2s+8}$$

$$\therefore 1700 \quad n3 \quad \dots$$

$\therefore$  we have  $s^3 + 2s^2 + 2s + 8$  to do the Routh table

$$s^3 \quad 1 \quad \quad \quad 2$$

$$s^2 \quad 2 \quad \quad \quad 8$$

$$s^1 \quad -\frac{\begin{vmatrix} 1 & 2 \\ 2 & 8 \end{vmatrix}}{2} \quad 0 \quad \therefore \text{Unstable}$$

$$= \frac{-(8-4)}{2}$$

$$= -2$$

$$s^0 \quad -\frac{\begin{vmatrix} 2 & 8 \\ -2 & 0 \end{vmatrix}}{-2} \quad 0$$

$$= -8$$

**19.** For the unity-feedback system of [Figure P6.3](#) with

$$G(s) = \frac{K(s+1)}{s^4(s+4)}$$

find the range of  $K$  for stability. [[Section: 6.4](#)]

Q19.

$$G(s) = \frac{K(s+1)}{s^4(s+4)} \quad \text{let } 1 + G(s)H(s) = 0 \rightarrow$$

$$\therefore s^4(s+4) + K(s+1) = 0 \rightarrow s^5 + 4s^4 + Ks + K = 0$$

$$s^5 \quad \quad \quad 1 \quad \quad \quad 0 \quad \quad \quad K$$

$$s^4 \quad \quad \quad 4 \quad \quad \quad 0 \quad \quad \quad K$$

$$s^3 \quad -\frac{\begin{vmatrix} 1 & 0 \\ 4 & 0 \end{vmatrix}}{4} \quad -\frac{\begin{vmatrix} 1 & K \\ 4 & K \end{vmatrix}}{4} \quad 0$$

$$= \frac{3K}{4}$$

$$s^2 \quad - \frac{\begin{vmatrix} 4 & 0 \\ \varepsilon & \frac{3k}{4} \end{vmatrix}}{\varepsilon} \quad - \frac{\begin{vmatrix} 4 & k \\ \varepsilon & 0 \end{vmatrix}}{\varepsilon} \quad 0$$

$$= \frac{-3k}{\varepsilon} \quad = k$$

$$s^1 \quad - \frac{\begin{vmatrix} \varepsilon & \frac{3k}{4} \\ \frac{3k}{\varepsilon} & k \end{vmatrix}}{\frac{-3k}{\varepsilon}} \quad 0 \quad 0$$

$$= \frac{4\varepsilon^2 + 9k}{12} \gtrsim \frac{3k}{4}$$

$$s^0 \quad - \frac{\begin{vmatrix} -3k/\varepsilon & k \\ 4\varepsilon^2 + 9k/12 & 0 \end{vmatrix}}{4\varepsilon^2 + 9k/12} \quad 0 \quad 0$$

$$= \frac{- (0 - k \cdot \frac{4\varepsilon^2 + 9k}{12})}{\frac{4\varepsilon^2 + 9k}{12}}$$

$$= k$$

$\therefore$  Assume  $\varepsilon > 0$  then let 1st column be all positive

$\therefore -3k > 0 \quad k < 0 \quad \therefore$  Cannot satisfied the stable

$$\frac{3k}{4} > 0 \quad k > 0$$

$$k > 0$$

if  $\varepsilon < 0$  then not stable

